

Single-cell transcriptional dynamics in a living vertebrate

Highlights

- Introduced technology for single-cell transcriptional dynamics in live zebrafish
- Improved the MS2-MCP system for fluorescent labeling of RNA in zebrafish
- Revealed transcriptional bursting behind segmentation clock oscillations
- Combined modeling and burst statistics to identify mechanisms of auto-repression

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In brief

Eck, Moretti, Schlomann, et al. established technology for quantifying single-cell transcriptional dynamics in live zebrafish embryos via imaging. Applied to the vertebrate segmentation clock, this approach revealed that the smooth oscillations of transcription factors orchestrating segmentation are generated by discrete transcriptional bursts ordered in space and time.

Methods

Single-cell transcriptional dynamics in a living vertebrate

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SUMMARY

The ability to follow transcription in individual cells with live imaging has revealed key dynamical mechanisms of gene regulation. However, such measurements are lacking in the context of vertebrate embryos. We addressed this deficit by applying MS2-MCP mRNA labeling to the quantification of transcription in zebrafish, a model vertebrate. We developed a platform of transgenic organisms, light-sheet fluorescence microscopy, and optimized image analysis that enables visualization and quantification of MS2 reporters. With these tools, we obtained single-cell, real-time measurements of the transcriptional dynamics of the segmentation clock. Our measurements reveal that smooth clock protein oscillations arise from discrete transcriptional bursts that are organized in space and time. Together, these results highlight how measuring single-cell transcriptional activity in the context of vertebrate organisms can reveal unexpected features of gene regulation and how this data can fuel the dialogue between theory and experiment.

INTRODUCTION

Development is driven by highly dynamic and coordinated gene regulatory programs. For example, rapid switches in Hox gene expression control limb development,^{1,2} transient excursions of gene expression that are subsequently refined create stripe patterns during body segmentation in fruit flies,³ and transcriptional oscillations direct cell fate decisions in both neural and segmental precursors.^{4,5} Innovations over the last few years have resulted in an explosion of our ability to map the gene regulatory networks dictating developmental dynamics. However, predictively understanding the links between the transcriptional dynamics, cell fate decisions, and tissue patterning encoded by these networks remains an open, fundamental problem in developmental biology.⁶

The MS2-MCP system is a widely used method for precise, quantitative measurements of real-time transcriptional dynamics of individual genes^{7–12} (Figure 1A). In this system, a gene is

tagged with multiple copies of a stem loop sequence from bacteriophage MS2 (referred to as MS2), whose cognate binding partner (MS2 coat protein, referred to as MCP) is fused to a fluorescent protein. Upon mRNA synthesis, MCP binds the MS2 loops within the nascent transcript, producing a region of enhanced fluorophore concentration (a “spot”) that can be detected with a fluorescence microscope (Figure 1A). The intensity of the resulting fluorescent spot is proportional to the number of RNA polymerase molecules actively transcribing the gene.

To date, precise, quantitative measurements of transcriptional dynamics via the MS2-MCP system have been limited to single cells in culture, tissue samples, invertebrate model organisms, such as flies and worms, and plants.^{11,13–18} In these contexts, the MS2-MCP system has revealed highly dynamic gene regulatory programs and molecular mechanisms of transcriptional regulation.¹⁹ However, similar measurements have yet to be achieved in a living, intact vertebrate. Given its established genetic toolkit and amenability to live imaging, the zebrafish is a

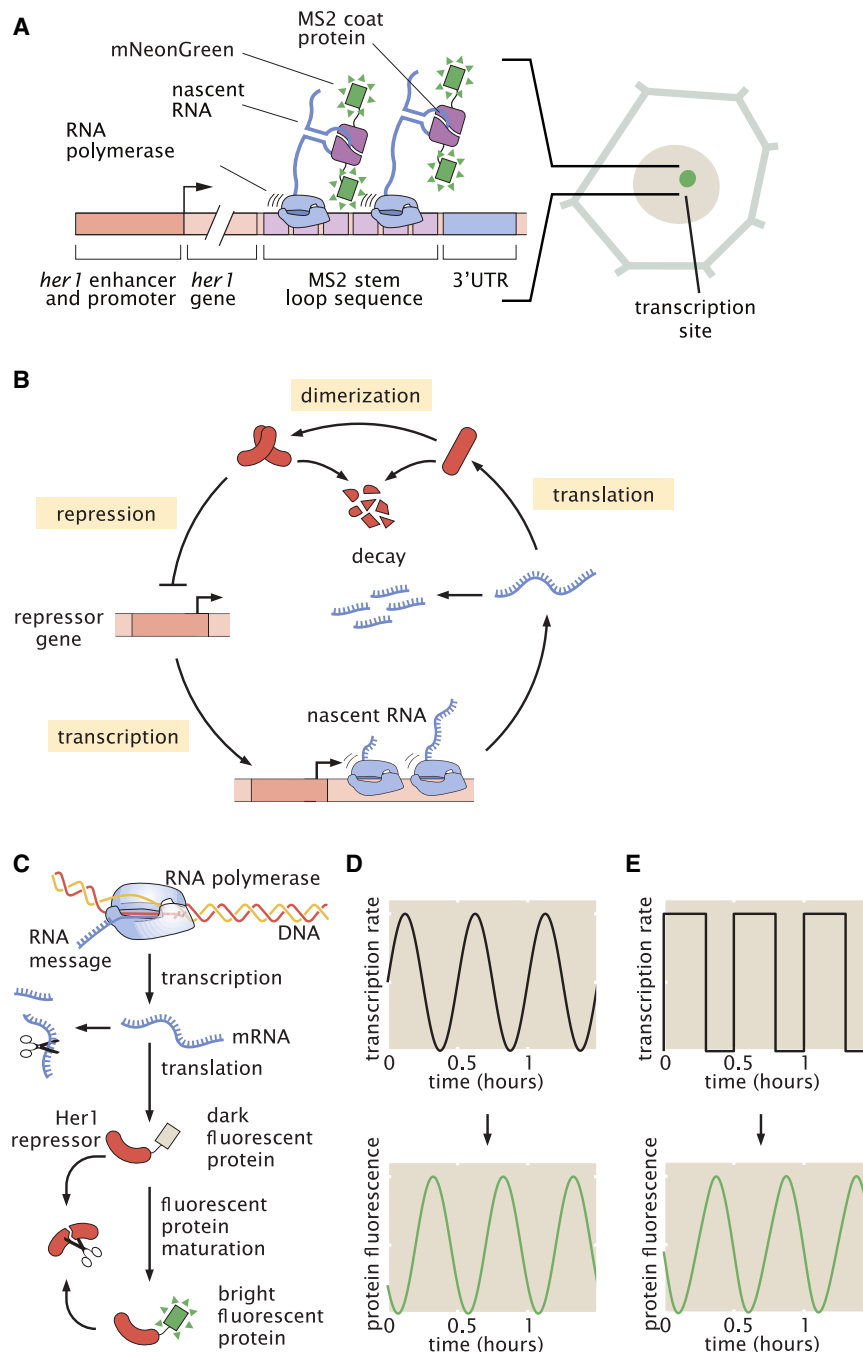


Figure 1. Protein reporters cannot distinguish between continuous and discrete modes of transcriptional regulation of the segmentation clock

(A) Schematic of the MS2 system, in which pre-formed fluorescent proteins (mNeonGreen) fused to MS2 coat protein bind to nascent mRNA strands via a stem loop-coat protein interaction. Fluorescence accumulates at sites of nascent transcript formation. In this way, the MS2 system gives a near-real-time readout of transcriptional activity. In this work, we tag a transgene containing the segmentation clock gene *her1* and its regulatory region.

(B) Oscillations are generated via auto-repression of segmentation clock genes such as *her1*.

(C) Schematic of the central dogma showing the steps that lead to low-pass filtering between transcriptional and protein dynamics.

(D and E) The low-pass filtering effect of product accumulation and decay leads to smooth protein oscillations that can be generated by dissimilar transcriptional dynamics ranging from smooth modulation (D) to discrete pulses (E).

both image acquisition and analysis. Due to fast cell motion during tailbud elongation (50 $\mu\text{m}/\text{h}$ is a typical speed; see ²⁴), fields of view hundreds of micrometers in length are required to track groups of cells for multiple hours. Simultaneously, one must be able to detect signals coming from diffraction-limited spots made of only tens to hundreds of fluorophores. Large fields of view, over time, result in large (here, ~ 1 TB) datasets. The large size of the data, together with dense nuclei and strong, heterogeneous fluorescence backgrounds, complicates cell tracking and MS2 spot quantification. Thus, the full power of the MS2 system to quantify transcriptional dynamics has yet to be realized in zebrafish or any other vertebrate animal for molecular biology, imaging, and data processing reasons. As many aspects of vertebrate development and physiology fundamentally differ from invertebrates, entire fields of questions

prime candidate for exploring single-cell transcriptional dynamics in vertebrates. While MS2 has been used in zebrafish to study mRNA localization, previous implementations have faced issues of MCP aggregation that prevented quantification of transcription.^{20–23}

Further, even if the genetic implementation of MS2 in vertebrates such as zebrafish was optimized, large challenges to using this technology to quantify transcriptional dynamics in intact vertebrate animals remain. For example, compared with flies or worms, zebrafish embryos are larger, denser, and exhibit greater motion during development, posing challenges in

concerning transcriptional dynamics—including questions of direct biomedical relevance—remain unanswerable.

Here, we report the establishment of the MS2 system in transgenic zebrafish together with the development of an allied platform of light-sheet fluorescence microscopy and computational image analysis to quantify transcriptional dynamics of MS2 reporters in complex and dynamical vertebrate embryos. To make this possible, we generated a new transgenic zebrafish line expressing a reengineered MCP that avoids protein aggregation. Further, to enable fast, large volume imaging, we used light-sheet fluorescence microscopy and demonstrated that

this method can retain the sensitivity required to detect subcellular MS2 spots while simultaneously imaging tens of thousands of cells in intact tissue. To enable single-cell tracking and spot quantification in large datasets with strong background, we developed a computational image analysis pipeline that combines parallelized chunk-based file handling, GPU-accelerated image processing, and deep learning, achieving fast and accurate measurements of transcriptional dynamics in single cells.

We applied this platform to a paradigmatic example of vertebrate-specific gene expression dynamics: the segmentation clock. In vertebrate embryo development, somites—morphological segments that prefigure the bones and muscles of the adult—are formed rhythmically and sequentially at the posterior end of the elongating body axis from the presomitic mesoderm.^{5,25} This rhythmic specification is dictated by a highly conserved biological clock consisting of an oscillatory gene regulatory network,²⁵ which is rooted in an auto-repression motif (Figure 1B). Protein reporters have revealed insights into the dynamics of these oscillators, revealing smooth, sinusoidal oscillations at both the tissue scale^{26–28} and in single cells.^{27,29–33} However, how oscillations are generated at the transcriptional level remains unknown. Due to the low-pass filtering effect of RNA and protein accumulation and degradation (Figure 1C), high-frequency dynamics are smoothed out at the protein level, making these transcriptional dynamics invisible to measurements afforded by the protein reporters available to date. For example, single-cell transcription rates could mirror protein levels, oscillating in a smooth, sinusoidal fashion (Figure 1D), or they could be sharp and discrete (Figure 1E). Both scenarios would produce smooth protein oscillations (see also Figure S1 for a mechanistic example based on smooth vs. sharp regulation of *her1* transcription rate).

At the tissue scale, our transcriptional measurements recapitulate the known global picture of somitogenesis: the expression of the core clock gene *her1* oscillates over time and travels in a wave-like pattern from the tailbud to the forming somites. However, at the single-cell level, we discovered that, in contrast to the smooth cycles of protein levels observed with fluorescent protein fusions in all vertebrate systems studied to date,^{29,33–36} transcription rate is modulated in a sharp, discrete manner, resulting in bursts of transcription. While these bursts are largely periodic, they also exhibit a degree of stochasticity. Simulations of the protein oscillations predicted to occur based on these transcriptional dynamics are regular and sinusoidal, indicating a mechanism of robust oscillations that is compatible with this stochastic, pulse-like transcription. To begin to explore how regular oscillations arise from transcriptional bursting, we developed a minimal mathematical model that extends the canonical two-state bursting model¹³ to include auto-repressive feedback. Analysis of waiting time distributions for the active and inactive transcriptional intervals provides evidence for a molecular mechanism of *her1* action consisting of either the regulation of burst amplitude or the regulation of a combination of frequency and duration.

Altogether, this work provides the tools necessary for studying transcriptional dynamics in zebrafish to the greater community and presents new measurements that challenge the existing paradigm of somitogenesis. Further, the pipeline presented here constitutes a framework that can be used to launch quanti-

tative live imaging and theoretical dissection of transcriptional dynamics in other vertebrate systems.

RESULTS

Implementing MS2 in zebrafish embryos to quantify transcriptional dynamics of the segmentation clock

While expression of transgenes via transient injection of one-cell-stage embryos is common in zebrafish,²⁰ this approach is not ideal for reproducible, quantitative measurements of transcription due to sources of variation such as mosaicism and positional effects.^{37,38} Therefore, we generated two stable transgenic zebrafish lines via I-SceI meganuclease transgenesis³⁹: one carrying a reporter construct containing MS2 stem loop sequences inserted downstream of the *her1* regulatory region, dubbed “*her1*-MS2,” and another carrying an MCP-mNeonGreen (herein MCP) fusion driven by a ubiquitin promoter⁴⁰ (Figures 2A and 2B; see the [zebrafish](#) section of the [STAR Methods](#)). The *her1*-MS2 construct was built off of a previously established *her1*-fluorescent protein fusion transgene construct exhibiting accurate oscillating gene expression patterns,^{26,33,41} with the fluorescent protein switched for the MS2 stem loops. Our approach differs from previous works on the MS2 system in zebrafish, which used transient injection of reporter constructs containing the MS2 loops.^{20–23}

We also made modifications to the MCP construct. Specifically, previous work used tandem dimer MCP with a nuclear localization signal²⁰ and saw unwanted aggregation of the coat protein. We suspected that this aggregation was due to tandem dimer MCP and therefore used regular MCP. Further, while the nuclear localization signal has proven useful for keeping background levels low when monitoring transcript localization in the cytoplasm,⁴² it can lead to overly high background levels in the nucleus that interfere with transcription measurements. Therefore, we did not include a nuclear localization signal in our construct. With these changes, our MCP line showed no aggregation, while the previous tandem dimer MCP line showed nuclear aggregates in the absence of an MS2 reporter (Figures S2A–S2D). Consistent with the lack of aggregation in our new MCP line, when crossed with our *her1*-MS2 reporter, 95% of cells contained only a single detected fluorescent spot corresponding to the single heterozygous copy of the *her1*-MS2 insertion (Figure S2E).

We used I-SceI transgenesis because it is highly efficient and typically generates single insertion events.³⁹ Single insertion events are ideal here because multiple insertion events could result in multiple MS2 spots, which would complicate our analyses. However, I-SceI transgenesis can insert multiple copies of the transgene at the same site.³⁹ Indeed, based on qPCR, we measured that our line carries 6 ± 1 copies of the *her1*-MS2 reporter ([STAR Methods](#)). These multiple copies raise two potential issues. First, multiple copies of the *her1* promoter might activate asynchronously, which would produce complicated MS2 signals. As we demonstrate below, we observe largely periodic signals, indicating that the multiple copies are acting in unison to a high degree. We elaborate on this point further in the [Discussion](#) section.

A second potential issue is that, since our construct contains the full *her1* coding sequence with a C-terminal fusion, it is likely

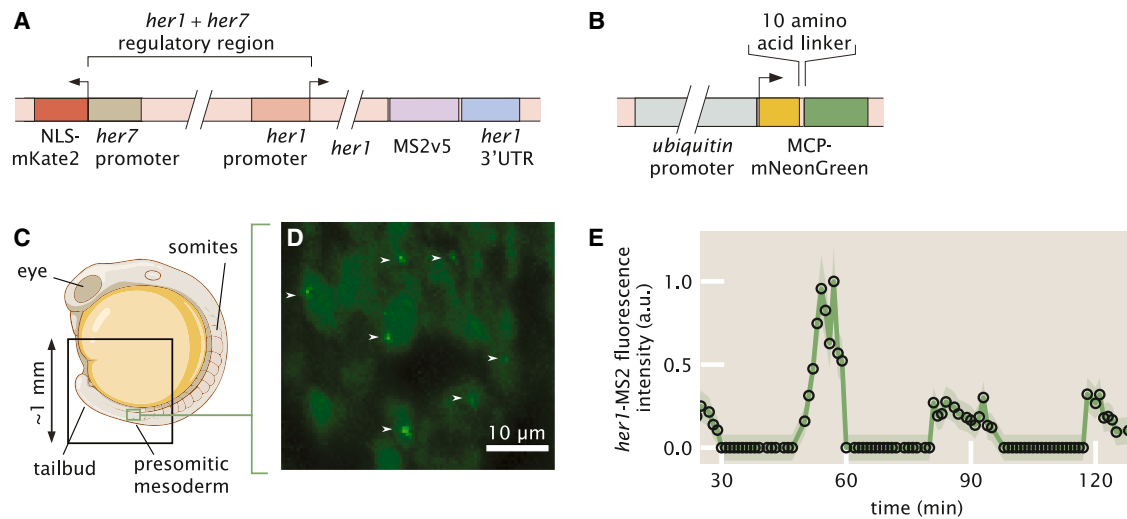


Figure 2. The MS2 system can be used to visualize transcriptional dynamics of the zebrafish segmentation clock

(A and B) Details of the transgenic MS2 (A) and MCP (B) constructs. Stable transgenic lines were made using I-SceI meganuclease transgenesis. The MS2 construct contains an mKate2 marker for ease of screening transgenic zebrafish.

(C) Geometry of a zebrafish embryo. Black box represents the approximate field of view of the imaging experiment. The green box represents the approximate field of view of the zoomed-in area shown in (D).

(D) Maximum intensity projection over 50 micrometers of the signal from MCP-mNeonGreen, *her1*-MS2 zebrafish showing several fluorescent spots (white arrows) against the hazy background MCP expression in individual nuclei.

(E) Example *her1*-MS2 trace obtained for an individual cell within the presomitic mesoderm showing oscillations. Error bars are reported by the shaded area and reflect uncertainty due to background fluctuations and spot detection (see the [spot analysis](#) section of the [STAR Methods](#)). (C) is adapted from BioRender.

that extra copies of Her1 are produced, potentially altering clock function and producing developmental defects. To address this issue, we performed extensive characterization of segmentation clock patterning and somite formation in our transgenic lines. For both MS2 and MCP zebrafish lines, heterozygous animals showed normal patterns of *her1* expression (Figures S3A–S3D) and minimal somite boundary formation defects consistent with previous Her1 protein reporter lines^{26,29,43,44} (Figures S3F and S3G), indicating healthy somitogenesis. Further, *in situ* hybridization against the MS2 sequence mirrored the patterns of *her1* expression (Figure S3E), demonstrating that our construct faithfully reports on *her1* activity.

To assess the feasibility of detecting *her1*-MS2 spots, we imaged embryos on multiple types of fluorescence microscopes. To visualize nuclei, we crossed MCP-mNeonGreen fish with a line carrying h2b-mScarlet⁴⁵ (see the [zebrafish](#) section of the [STAR Methods](#)). In *her1*-MS2/+; MCP-mNeonGreen/+; h2b-mScarlet/+ animals, green fluorescent puncta representing nascent *her1*-MS2 transcripts were readily detectable using both light sheet (Figures 2C and 2D; Figures S4A and S4B) and laser-scanning confocal (Figures S4C and S4D) fluorescence microscopes. In control experiments with embryos expressing only MCP-mNeonGreen and not *her1*-MS2, fluorescent puncta were not detected (Figure S4B).

In initial time-lapse imaging experiments on a laser-scanning confocal (Zeiss 880 with AiryScan), we found that achieving even a barely detectable MS2 signal with 1 min time resolution (required to capture substructure within the oscillations, whose period is ~30 min²⁹) resulted in a small field of view of $100 \times 50 \times 20 \mu\text{m}^3$, or approximately 50 nuclei. To image the

full tissue would require a field of view of approximately $500 \times 500 \times 400 \mu\text{m}^3$. Due to significant cell motion during somitogenesis,²⁴ we found that manual adjustment of the microscope stage was required every few minutes to keep the same cells in the field of view (Video S1). To capture multiple oscillations, this constant adjustment must be done for multiple hours and the resulting Videos registered together, making this approach impractical. Consequently, we exclusively used light-sheet fluorescence microscopy, which enabled us to capture the full tissue containing somites, presomitic mesoderm, and tailbud in a $1,000 \times 1,000 \times 388 \mu\text{m}^3$ field of view with 1 min time resolution and acceptable signal-to-noise ratios⁴⁶ (see the [light-sheet imaging](#) section of the [STAR Methods](#)).

We built a computational image analysis pipeline (Figures S5A and S5B) to extract quantitative measurements of transcriptional dynamics from our images. As elaborated in the [spot analysis](#) section of the [STAR Methods](#), the large size (~1 TB) of the image datasets and the strong and heterogeneous background signal posed new challenges to the analysis of MS2 data. Specifically, the large size of the data called for out-of-memory computing schemes and speed optimization of every step of the pipeline, while the strong background required sophisticated algorithms for spot classification and localization. Overall, these challenges, which were new to the analysis of MS2 data in general, necessitated the development of specialized software. Our pipeline tracks nuclei and identifies spots in parallel, then spots are assigned to nuclei, and the sequences of MS2 spot intensities over time—“traces”—are constructed. Initial examination of traces revealed clear oscillations (Figure 2E).

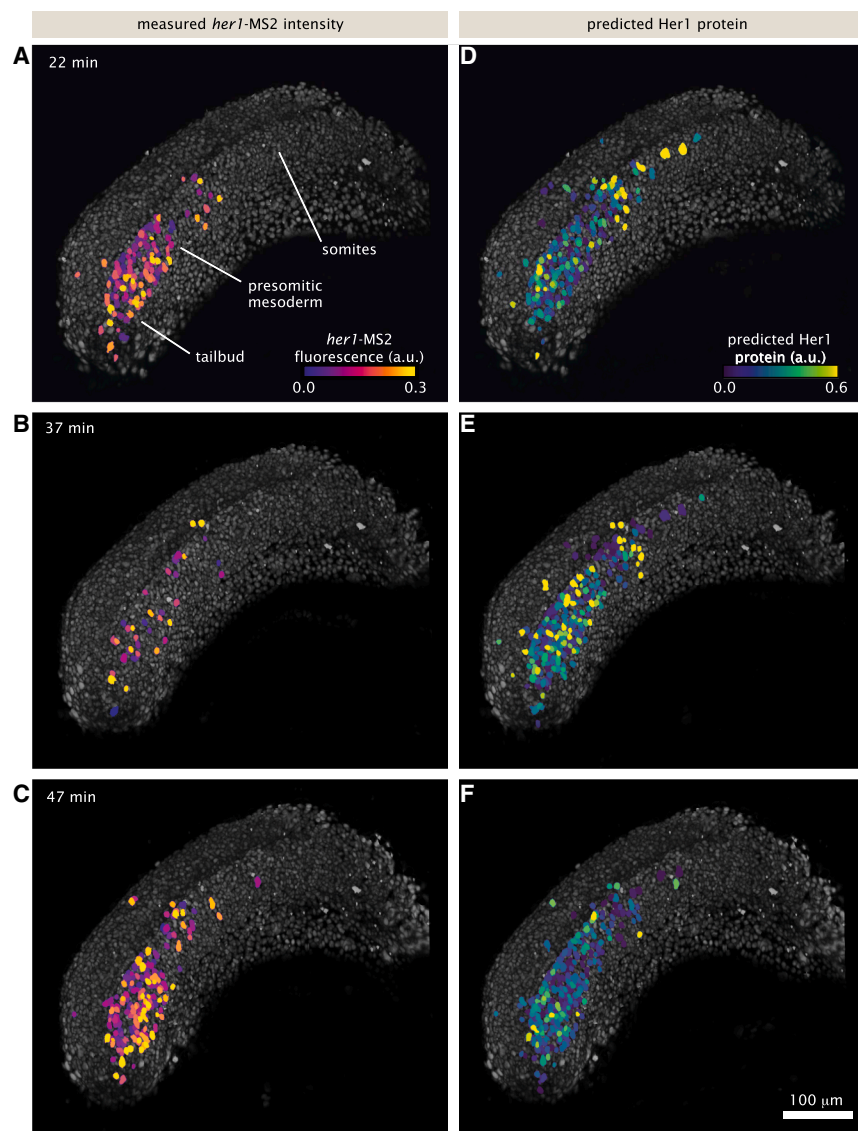


Figure 3. Transcriptional dynamics at the tissue scale recapitulate known features of protein patterns

Snapshots of 3D renderings of the presomitic mesoderm and tailbud during somitogenesis (A–C) Nuclei (gray) are false-colored in proportion to the intensity of their MS2 spots to achieve visual clarity.

(D–F) Same as (A)–(C), but nuclei are colored in proportion to the predicted Her1 protein level, given the MS2 signal in (A)–(C) (STAR Methods). Both MS2 and predicted protein intensities were normalized to their respective maximum values across the Video. See also Videos S4, S5, and S6. Scale bar, 100 μ m. Time is measured from the start of image acquisition, which corresponds to approximately the 15 somite stage.

analysis of synthetic images containing simulated spots, we found that our spot fluorescence quantification algorithm was also highly accurate, with typical errors of around 5% (Figure S6; STAR Methods). As our MCP background was expressed quite heterogeneously throughout the tissue, it is possible that in some cells MCP levels were low enough to not saturate the MS2 stem loops, introducing a background-dependent signal⁴⁷ (Figure S7). However, we observed only a very weak correlation ($R^2 = 0.26$) between the fluorescence background of spots and their (background subtracted) signal (Figure S5E), indicative of MCP saturation and reliable measurements of absolute fluorescence intensity.

Taken together, these results demonstrate that we successfully implemented the MS2 system in zebrafish for quantifying

We performed extensive characterization of the pipeline’s accuracy (see the [spot analysis](#) section of the STAR Methods; Figures S5C–S5H). Through comparison with 26 manually obtained traces, we found that our automated 3D nuclear tracking was approximately 80% accurate (21/26 nuclear trajectories showed no errors), with an error rate of around 1 nuclear tracking error per 5 nuclear trajectories, or on average 1 error per 500 min (see also Video S2 and S3). Examples of tracking errors include swapping nuclear identities between neighboring cells, grouping two nuclei into one, and splitting one nucleus into two (Figure S5G). These errors were manually corrected.

Through comparison with manually curated traces, where spot detection was done entirely by human visual inspection, we found that spot detection was extremely accurate. Specifically, we found a false positive rate of 4% and a false negative rate of 8% (Figures S5H and S5I). When both manual traces and pipeline traces detected a spot, the resulting spot intensities were highly correlated with an R^2 of 0.91 (Figure S5I). Through

ifying *her1* transcriptional dynamics. We then turned to studying the behavior of *her1* transcription at tissue and single-cell scales.

***her1* transcriptional dynamics recapitulate protein oscillations and wave patterns at the tissue scale**

We first assessed how the tissue-scale dynamics of the segmentation clock at the transcriptional level compare to the measurements obtained from Her1/Her7 protein dynamics reported by fluorescent protein fusions.^{26–33} Previous protein measurements revealed collective oscillations whose period increases as cells approach somite formation, resulting in a phase wave that travels from posterior to anterior, arresting at the formation of a new somite.^{26,29} The oscillation period varies with temperature, being approximately 30 min at 28 C.

While the dynamics of protein reporters can be directly visualized in movies,^{26,29} MS2 spots are too small and dim to be picked up in raw 3D renderings or projections of the full tissue. Therefore, to visualize transcriptional dynamics, we false-colored nuclei in proportion to their spot intensity (Figures 3A–4C; Videos S4 and

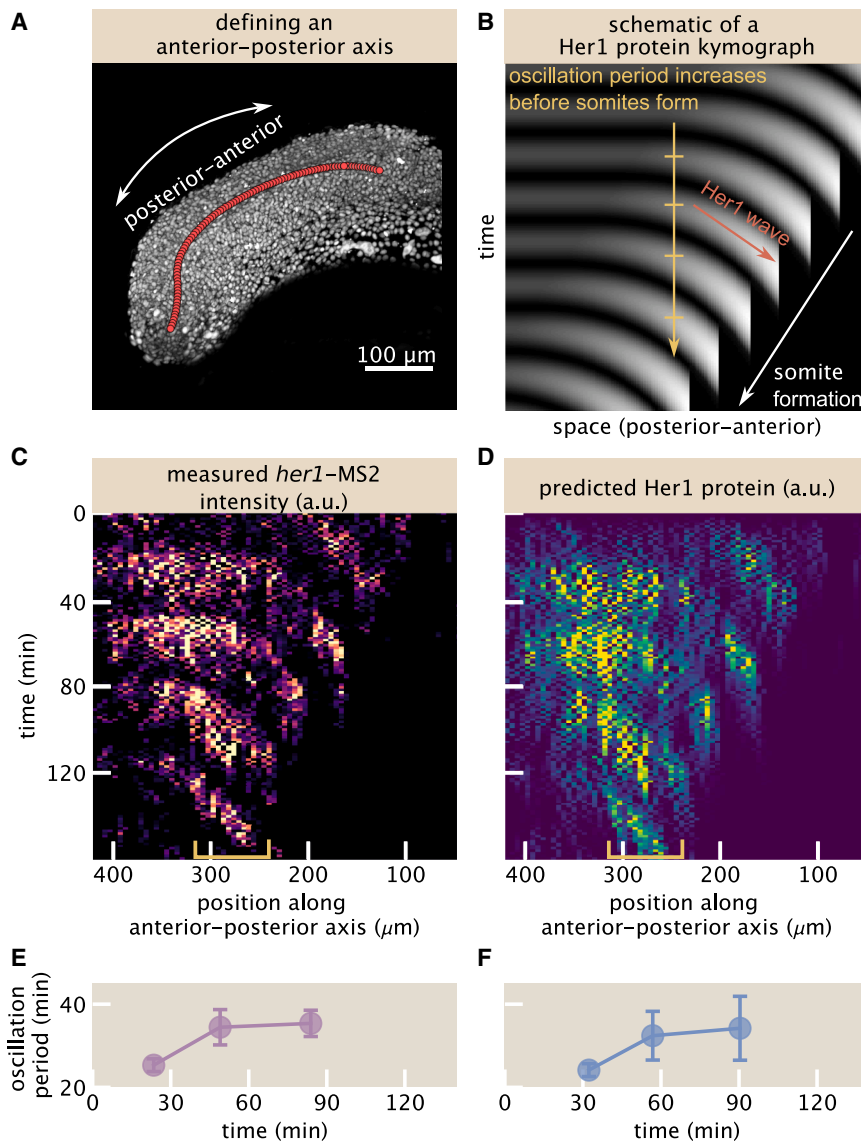


Figure 4. Tissue-scale analysis reveals transcriptional oscillations and wave-like patterns

(A) 3D rendering of nuclei (gray) illustrating the anterior-posterior axis (red circles). The axis is defined coarsely every 50 time points, roughly following the notochord, and then spline interpolated in both space and time (see the [anterior-posterior axis](#) section of the [STAR Methods](#)). See also [Video S7](#).

(B) Schematic of a Her1 protein kymograph highlighting the known features. Oscillation period increases as cells approach somite formation, which produces a traveling wave pattern.

(C and D) Kymographs of total *her1*-MS2 (C) spot intensity and predicted Her1 protein (D). Distance is defined along the anterior-posterior axis, as in (A). Spot anterior-posterior positions are binned, with bin size = $6.5 \mu\text{m}$, and the sum of all intensities in each bin is plotted for each time point.

(E and F) Measurement of the increase in oscillation period as cells approach somite formation for the raw *her1*-MS2 kymograph (E) and the predicted Her1 protein kymograph (F). The regions used to compute the period are noted by the yellow brackets in (C) and (D).

given by m and p , respectively. The model considers our MS2 traces as a proxy for the rate of transcription, $r_m(t)$, as input,⁴⁸ and has parameters for the mRNA decay rate (γ_m) and protein decay rate (γ_p ; see the [protein signal prediction](#) section of the [STAR Methods](#) for more details and parameter choices). Since here we are only concerned with relative, not absolute, changes in protein concentration, we choose units such that r_p is unity. With this model, we generated visualizations of the predicted Her1 protein concentrations ([Figures 3D–3F](#); [Video S6](#)). The predicted protein patterns are

[S5](#)). The resulting Videos showed oscillations and waves similar to movies of fluorescent proteins, qualitatively recapitulating previous protein-level dynamic measurements, albeit with a higher prevalence of noise and high-frequency flashing.

To validate our construct more directly against previous protein measurements, we compared the protein dynamics predicted by our transcriptional data to actual protein-level measurements of Her1 dynamics. We used a linear model of the central dogma given by

$$\frac{dm}{dt} = r_m(t) - \gamma_m m$$

and

$$\frac{dp}{dt} = r_p m - \gamma_p p$$

(based on [Figure 1C](#)). This model predicts the dynamics of mRNA and protein concentrations, measured in arbitrary units,

smoother than the MS2 patterns, showing gradual waves traveling from posterior to anterior that resemble previous measurements.^{26,29} Thus, once protein production and decay are taken into account, our transcriptional reporter faithfully recapitulates the behavior of established protein reporters at a qualitative level.

To quantitatively validate our reporter, we captured transcription and protein spatiotemporal patterns in kymographs ([Figure 4](#)). To make this possible, we defined an anterior-posterior axis with a combination of manual labeling and spline interpolation ([Figure 4A](#); [Video S7](#); [STAR Methods](#)). We grouped cells into equally spaced bins along this axis, summed the spot intensities in each bin, and plotted them over time. Based on previous measurements,²⁶ we expected the kymographs to resemble the schematic in [Figure 4B](#) (see the [kymograph creation and tissue-scale period measurement](#) section of the [STAR Methods](#)), which we briefly describe here. In these kymographs, with time increasing downward and anterior to the right, horizontal stripes

correspond to the stationary oscillations that occur in the tailbud, while diagonal stripes correspond to waves that travel toward the anterior and arrest as somites form. If one picks a point in absolute space and follows the oscillations over time, the oscillation period increases. We hypothesized that these features, which were identified in kymographs of Her1 protein,²⁶ should be apparent in the kymograph of *her1* transcriptional dynamics.

The resulting MS2 kymograph (Figure 4C) shows clear, regular oscillations throughout the tissue, with periods of approximately 30 min, consistent with protein measurements.²⁶ Further, the left-down slope of the stripes indicates the traveling phase wave from posterior to anterior, with a speed of approximately 100 $\mu\text{m}/\text{h}$, also consistent with previous measurements.²⁶ The predicted protein kymograph (Figure 4D) largely mirrors the MS2 kymograph but with a phase shift, as expected from the time lag between transcription and translation. Quantitatively, we observed an increase in the oscillation period in the kymograph of approximately 40% prior to somite formation for both MS2 and predicted protein signals (Figures 4E and 4F; Figure S8).

All together, these observations demonstrate that, at the tissue scale, the transcriptional dynamics of the segmentation clock largely mirror protein dynamics, albeit with transcriptional activity appearing noisier than accumulated protein levels.

Single-cell transcriptional oscillations comprise sharp, quasi-periodic bursts

Exploiting the unique power of the MS2 system, we next investigated transcriptional dynamics in single cells in a set of 101 traces that were manually checked and corrected as needed. Individual MS2 traces (Figure 5A, green lines; Figure S9) show clear periodicity. However, in contrast to previously measured smooth protein oscillations,^{26,29,33} our observed transcriptional dynamics are sharp and discrete. Further, we observed examples of multiple, distinct transcriptional pulses in rapid succession, which we refer to as bursts (Figure 5A, black arrows). We visually inspected these bursts and confirmed the rapid disappearance and reappearance of the MS2 spot (Figures 5B and 5C). Even for these cases with a higher frequency of transcriptional bursting, predicted protein traces for single cells are smooth, roughly sinusoidal, and regular in period, similar to measured protein traces (Figure 5A, cyan lines; STAR Methods).^{29,33} With a conservative burst-calling algorithm (STAR Methods), we measured that $8\% \pm 3\%$ (mean $\pm$ std. dev. from bootstrapping over cells) of protein oscillations are generated by 2 or more transcriptional bursts (Figure 5E; STAR Methods). Comparing the periods of transcriptional bursts and the predicted protein oscillations, we found that the predicted protein oscillations were more regular than the transcriptional bursts, having a narrower, more peaked distribution (Figure 5F). Thus, our measurements of transcriptional bursts in the segmentation clock are consistent with the observation of regular protein oscillations, even without invoking additional levels of post-transcriptional regulation.

These sharp, noisy pulses are reminiscent of transcriptional bursts in other unicellular and multicellular systems.^{13,15,49–52} Canonical bursts are stochastic,^{13,48} at odds with the notion of regular oscillations. Indeed, previous analyses of noise metrics in single-molecule fluorescence *in situ* hybridization (smFISH)

data suggested that *her1* is transcribed in stochastic bursts,^{28,53} and in reanalyzing these data, we found explicitly that the distribution of mRNA *her1* counts is well-fit by the canonical two-state model of transcriptional bursting (Figure S10). However, unlike canonical bursty systems, the segmentation clock is strongly regulated by auto-repression: Her1 proteins repress *her1* transcription (Figure 1B). This feedback loop clearly could produce or modulate bursts in such a way as to generate regular oscillations. To shed light on the mechanism by which Her1 regulates its own transcriptional bursting and how stochastic transcriptional bursts might interplay with auto-repression to generate noisy but regular oscillations, we turned to mathematical modeling.

Burst timing statistics offer a window into molecular mechanisms of repression

We extended the classic two-state bursting model to include feedback that accounts for the auto-repression of *her1* (Figure 6A), ignoring interactions with other clock components such as *her7* (Method S3). Feedback can occur via regulation of burst amplitude (“amplitude regulation”) (Figure 6Ci), burst separation (“separation regulation”) (Figure 6Cii), burst duration (“duration regulation”) (Figure 6Ciii), or a combination thereof (Figure 6Civ). We found that quasi-regular oscillations can emerge from the regulation of either amplitude, separation, or duration (Figure S11). Since all three aspects of bursting dynamics are valid knobs for creating oscillations, dynamically rich regulatory strategies in which multiple aspects of bursts are modulated are possible, calling for theoretical predictions that can be tested to rule in or out each strategy.

As a first step in uncovering the mechanisms regulating *her1* bursts, we sought to identify signatures of the different regulatory strategies of bursting dynamics in the distribution of burst durations and separations (Figure 6B). The burst duration is the length of time that the promoter is in the ON state. The burst separation is the interval over which the promoter is in the OFF state. In an unregulated two-state model, the distributions of both intervals are exponential, reflecting the Poisson nature of the ON and OFF switching events. At the opposite extreme, regular oscillations have narrow, peaked interval distributions that become more peaked as the oscillations become perfect. We reasoned that when k_{on} is regulated but not k_{off} , the burst duration (controlled by k_{off}) will follow an exponential distribution while the burst separation (controlled by k_{on}) will be peaked to allow for bursts to occur in a periodic fashion. Alternatively, for k_{off} regulation, the burst separation will be exponentially distributed, while the burst duration will be peaked.

To assess whether these differences in the shape of burst interval distributions could be used in practice to distinguish between mechanisms of burst regulation, we measured the burst duration and separation distributions in stochastic simulations that included a model of the MS2 measurement process, choosing model parameters based on the literature where possible (see the generalized bursting model with feedback section of the STAR Methods). As expected, models that produce quasi-regular oscillations (Figure S11) using only separation or duration regulation have one peaked and one monotonic interval distribution (Figures 6Dii, 6Diii, 6Eii, and 6Eiii). These results are robust to a wide range of simulated measurement noise

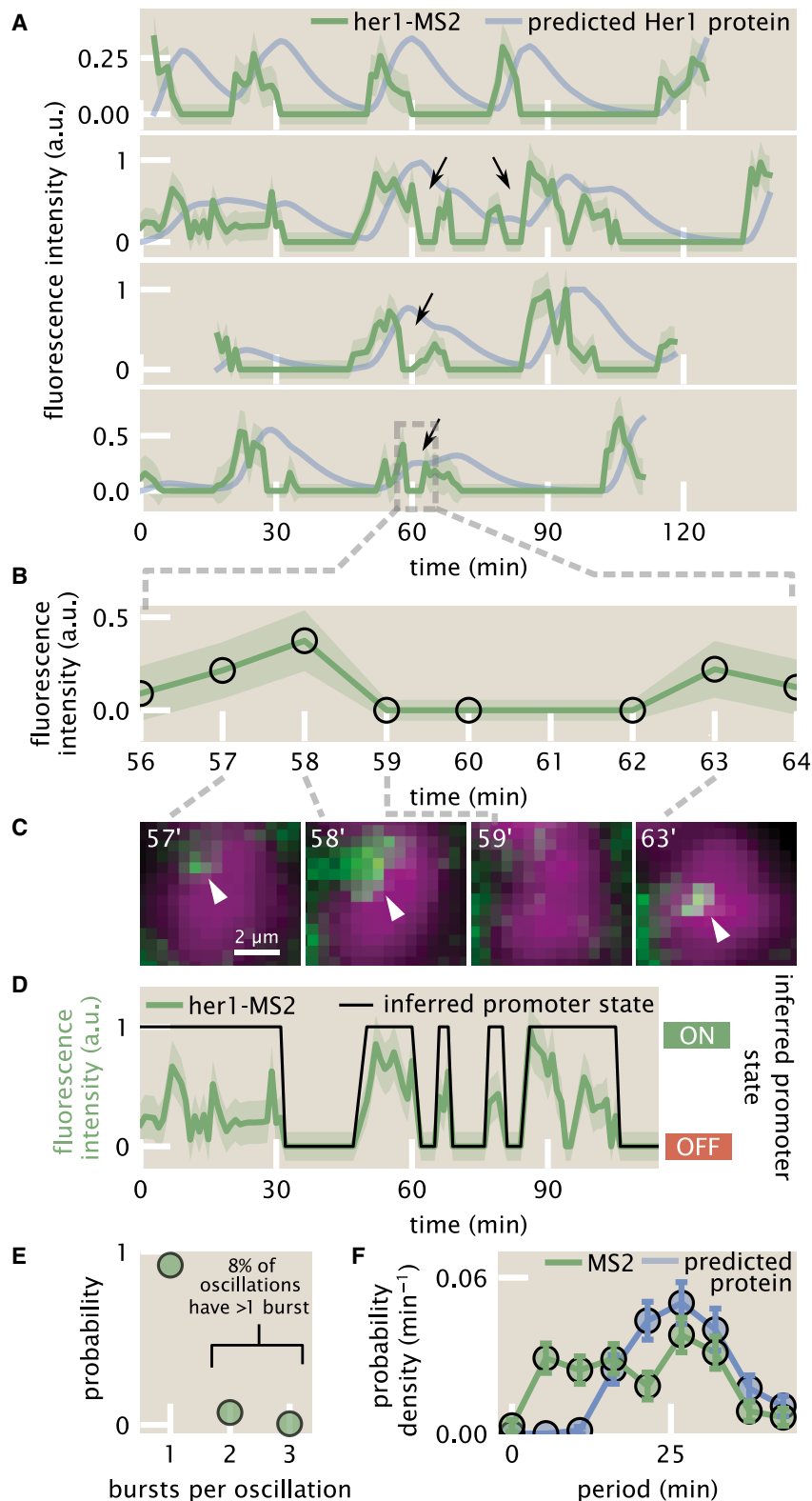


Figure 5. Single-cell traces reveal transcriptional bursts

(A) Measured single-cell *her1*-MS2 traces (green) and predicted Her1 protein levels (blue) using the production/decay model from Figure 1C (see the [protein signal prediction](#) section of the STAR Methods). Examples of multiple bursts giving rise to a single predicted protein oscillation are highlighted with black arrows. Shaded error bars reflect uncertainty due to background fluctuations and spot detection (see the [spot intensity uncertainty](#) section of the STAR Methods).

(B) Zoom in to one burst in the last trace of (A).

(C) Single z-slices images showing examples of the rapid disappearance and reappearance of a *her1*-MS2 spot, corresponding to the burst highlighted in (B). The z-slice of min 59 is the same as the previous time point. Spots are highlighted with white arrowheads. Green, *her1*-MS2; MCP-mNG. Magenta, h2b-mScarlet.

(D) Example of promoter state inference via trace binarization.

(E) Distribution of number of bursts per predicted protein oscillation for two embryos. $8\% \pm 3\%$ of oscillations have multiple bursts within them.

(F) Distribution of oscillation periods for *her1*-MS2 traces (green) and the predicted Her1 protein signal (blue). Protein oscillations are predicted to be more regular than the measured transcriptional oscillations as revealed by the more peaked distribution of protein periods. Error bars denote standard deviations over bootstrapped distributions.

contributions for both burst duration and separation (Figures 6Di, 6Div, 6Ei, and 6Eiv). Note that in the case of amplitude regulation, the auto-repression itself creates effective “ON” and “OFF” promoter states that are measured by MS2 (Figure S11). Therefore, these differing predictions for burst timing statistics offer an opportunity to uncover which regulatory strategies are at play, even in the presence of realistic measurement noise.

Turning to the data, we found peaked distributions for both burst duration and separation (Figures 6Div and 6Eiv). This result rules against a model of purely separation or duration regulation but calls for either amplitude regulation, a combination of separation and duration regulation, or a combination of amplitude regulation with either or both of the separation and frequency regulations (see Figures S16 and S17 for analysis of all combinations of regulatory strategies). To further constrain these models, we looked for changes in burst parameters within single cells as

parameters (Figures S12–S15). By contrast, simulated *her1*-MS2 signals from models that combine separation and duration regulation or have only amplitude regulation produce peaked distri-

they approach somite formation, during which protein oscillations increase in period.³³ Despite our *her1*-MS2 measurements showing a period increase at the tissue scale (Figure 4E), we

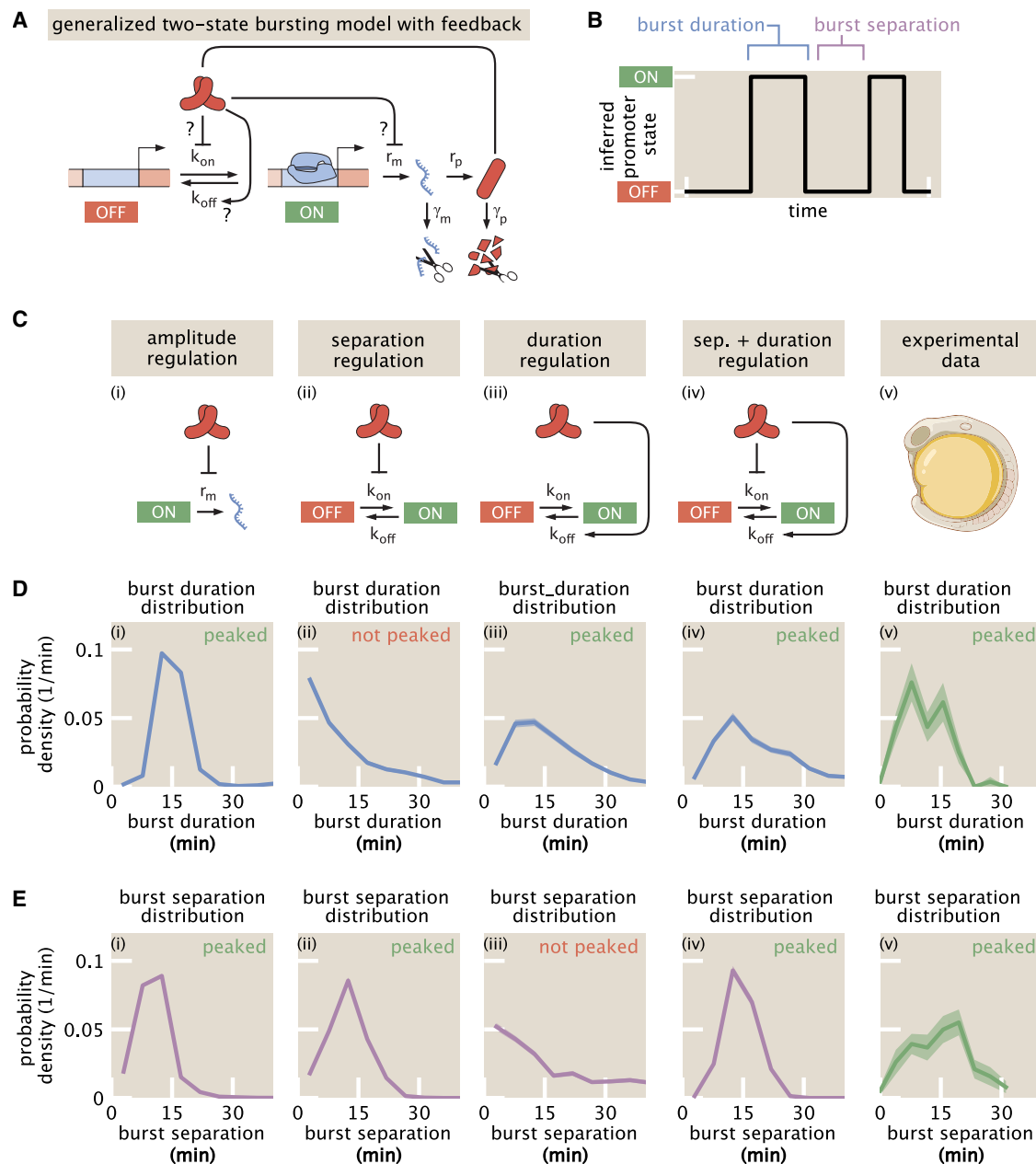


Figure 6. A two-state model of transcriptional bursting with feedback suggests multiple mechanisms of repression could generate oscillations

(A) Schematic of the bursting model. The promoter switches between active (ON) and inactive (OFF) states with rates k_{on} and k_{off} , respectively. In the active state, mRNA is transcribed at rate r_m and then translated into protein.

(B) Schematic of the duration and separation of bursts.

(C) The protein can modulate any of r_m , k_{on} , and k_{off} to repress mRNA production and generate oscillations, which we term (i) amplitude, (ii) separation, and (iii) duration regulation, respectively. We also consider the case of combined separation and duration regulation (iv). See [Figures S16](#) and [S17](#) for other combinations. Our goal is to find evidence for or against these mechanisms in our experimental data from the zebrafish embryo (v).

(D and E) Simulated (i–iv) and measured (v) duration (D) and separation (E) distributions, with simulations covering different mechanisms of repression and including a simulated MS2 reporter with measurement noise (see the section [generalized bursting model with feedback](#) in the [STAR Methods](#)). Only models with amplitude regulation or combined separation and duration regulation produce peaked distributions for both burst duration and separation, as seen in the data. When only separation or duration is regulated, one of the interval distributions becomes exponential, even accounting for a wide range of measurement noise parameters ([Figures S12–S15](#)). For the measured distributions, shaded error bars represent standard deviations across bootstrapped distributions.

observed no changes in single-cell burst parameters over this time (Figure S18). However, given the strong degree of stochasticity in the bursts, our limited statistical power may be masking trends in burst parameters along the anterior-posterior axis that more data would reveal.

DISCUSSION

In this work, we established the MS2-MCP mRNA labeling system for quantifying transcriptional dynamics in zebrafish embryos, to our knowledge, a first in any intact vertebrate animal. To facilitate reproducible and quantitative measurements of RNA polymerase activity, we generated a stable line of *ubiquitin*-driven MCP-mNeonGreen that remedies aggregation issues seen in previous lines²⁰ and avoids many of the issues that accompany approaches in which the transgene is transiently injected for each experiment.³⁸ This line will aid the expansion of transcriptional dynamics research throughout the zebrafish community, as it can be combined with any MS2 construct. To robustly extract single-cell transcriptional dynamics across a millimeter of highly motile tissue (the elongating tailbud), we developed a platform of light-sheet fluorescence microscopy and custom image analysis routines that can process terabyte-scale datasets in a few hours. We used this technology to measure the dynamics of *her1*, a genetic oscillator in the core of the segmentation clock. In contrast with the smooth, sinusoidal oscillations seen at the protein level,^{26,29} at the transcriptional level oscillations are discrete and burst-like. Due to the low-pass filtering effects of RNA and protein accumulation, these high-frequency bursting dynamics get smoothed out at the protein level.⁵⁴ Thus, our finding highlights the power of the MS2 system in being able to reveal dynamical features of gene regulation that would remain invisible to classic approaches relying on fluorescent protein fusions.

Our observation of segmentation clock oscillations being composed of discrete transcriptional bursts raises the question of how these bursts are created and regulated to ensure proper protein dynamics that will ultimately drive vertebrate segmentation. In a simple mathematical model, we found that quasi-regular oscillations could be produced if Her1 protein modulated burst amplitude, separation, or duration, or a combination thereof. This finding reveals a much richer landscape of potential regulatory mechanisms that could be involved in the transcriptional control of the segmentation clock than had been previously considered.^{55–57} Studying burst timing distributions provided evidence in favor of either a mechanism of amplitude regulation, joint separation and duration regulation, or amplitude regulation with either or both of separation and duration regulation and provided evidence against a model of purely separation or duration regulation. However, experiments that perturb the negative feedback loop are required to verify these inferences. In future work, we will directly identify the mechanism of regulation by measuring *her1* transcription while knocking down Her1 protein, either through mutants or morpholinos, and looking for changes in burst amplitude, separation, or duration. Such an experiment is only possible with RNA labeling technologies like MS2: as demonstrated in Figures 1D and 1E, these different mechanisms can give rise to identical protein dynamics and are only distinguishable at the level of transcrip-

tional dynamics. Our tools thus present a unique opportunity to dissect the transcriptional regulation of an auto-repressive system.

In characterizing the relationship between transcriptional bursts of *her1* and the predicted Her1 protein oscillation, we found examples of multiple bursting events within one protein cycle. This observation challenges the simple picture of *her1* transcription rate being smoothly modulated by approximately sinusoidal Her1 protein oscillations⁵⁸ but rather points to the existence of more complicated and faster transcriptional dynamics. However, one potential caveat to this result is the possibility that the 6 copies of the *her1*-MS2 reporter construct inserted during transgenesis are activating independently and asynchronously. The fact that the majority of *her1*-MS2 traces are regular, with one transcriptional burst per protein oscillation, suggests that the 6 copies are strongly coupled such that they act largely in unison. This interpretation is consistent with recent work on the pairing of *her1* and the adjacent clock gene *her7*, which showed that when *her1* and *her7* are on the same chromosome (in *cis*), they are more likely to transcribe concurrently than when they are on opposite chromosomes (in *trans*).²⁸ For future iterations of the *her1*-MS2 reporter, new landing site technologies for the insertion of transgenes in a single copy would make it possible to circumvent any of the potentially confounding effects of multiple copy insertions.^{59–62}

Regardless, most of our results are largely insensitive to the reporter copy number. The tissue-scale dynamics of Figure 4C average over individual cells, washing out small-scale fluctuations that could result from asynchronous firing of multiple reporter copies. The distributions of active and inactive transcriptional intervals in Figures 6D and 6E would be sensitive to different copies of the promoter firing asynchronously if asynchronous bursts were frequent. However, the fraction of multiple burst events per protein oscillation is small (~8%). Therefore, even if we assume that all of these multiple burst events are the result of the individual and asynchronous firing of multiple copies of the reporter and exclude them, the overall shape of the burst duration and separation distributions would be unaffected. Finally, the measurement most affected by multiple copies would be the distribution of the number of bursts per protein oscillation (Figure 5E).

One further limitation of our results is that, at present, it is unclear to what extent one can compare the amplitude of our MS2 signals from different nuclei. Systematic error in the overall fluorescence intensity of spots can arise from multiple sources, including cell-cell variability in MCP expression from our ubiquitin promoter, variable light scattering with tissue depth, and non-uniformity of the excitation beam in light-sheet imaging. Exploring alternative promoters with less variability for driving MCP expression will be a useful future endeavor and will also likely improve spot detection (Figure S2). Further, while the quantification of fluorescence with laser-scanning confocal microscopy is relatively well established, future work is needed to more rigorously assess how to maximize the accuracy of *in vivo* quantitative fluorescence measurements done with light-sheet microscopy. Addressing these issues will be important for inferring the degree of burst amplitude regulation present in the segmentation clock, though it should not affect inference of burst timing regulation.

As a final limitation of our reporter, we note that some uncertainty remains regarding the extent to which our measurements recapitulate the single-cell transcriptional dynamics of the endogenous *her1* gene. Analysis of previously published smFISH data on endogenous *her1* transcripts strongly suggests that the endogenous gene exhibits bursting dynamics (Figure S10). However, future work tagging the endogenous *her1* locus is required to unambiguously resolve quantitative differences in bursting kinetics between our current reporter and the endogenous gene.

Outside the segmentation clock, the role of transcriptional bursting in genetic oscillators is relevant to a wide range of phenomena. While most eukaryotic genes are thought to be transcribed in bursts,^{13,63,64} the extent to which bursting is suppressed in oscillatory genes to enhance their timekeeping ability is only beginning to be revealed. We are aware of three other examples of MS2 measurements of oscillating genes, two that report bursts within protein-level oscillations and one that does not. Hafner et al. measured transcription in the p53 pathway and found that approximately 30% of p21 oscillations (period ~5 hours) contained multiple bursts.⁶⁵ Wildner et al. studied transcription in the yeast gene CUP1 in response to metal stressors and inferred bursting on the timescale of minutes within oscillations of period ~30 min.¹⁴ Kinney et al. observed oscillations in a gene involved in the circadian rhythm of *C. elegans* and found smooth modulations in transcription rate without burst-like dynamics.⁶⁶ Together with our results, these findings suggest that transcriptional bursting is a common, though not universal, feature of genetic oscillators and that suppressing bursting is not necessary to create regular oscillations.

Beyond zebrafish, we expect several of the challenges faced here to also be present in other vertebrate systems, such as organoids and mouse pre-implantation embryos. The large size and photosensitivity of these systems make light-sheet fluorescence microscopy a useful imaging approach, which comes with well-known challenges of image processing and analysis. Further, these tissues are also often dense and possess strong, structured fluorescence backgrounds that complicate particle detection.⁶⁷ Our image analysis pipeline for MS2 spot quantification, along with the plasmids for MCP and MS2 constructs, may be a useful starting point for such endeavors.

In sum, we envision the tools presented here—in combination with theoretical models—will be useful for expanding the field of quantitative developmental biology further into the domain of vertebrate tissues, where they will help us understand how highly dynamic gene expression programs achieve robustness and how they fail to do so in disease states.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Hernan G. Garcia (hggarcia@berkeley.edu).

Materials availability

Plasmids and transgenic zebrafish lines generated in this study (the *her1*-MS2 reporter line and the *ubi*:MCP-mNeonGreen line) are available from the [lead contact](#) upon request. Sequences for all constructs used in this work can be found at https://benchling.com/garcialab/f/_f9Cist7M-zebrafish-ms2-paper/.

Data and code availability

- Raw microscopy images are available upon request. Manually curated single-cell traces and supporting tabular data are also provided as [supplemental information](#) files.
- All original code has been deposited at GitHub (<https://github.com/GarciaLab/zms2> for the image analysis pipeline and <https://github.com/GarciaLab/zebrafish-ms2-paper> for the analysis and modeling code used to generate the figures) and archived at Zenodo. DOIs are listed in the [key resources table](#).
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization: E.E., B. Moretti, B.H.S., A.C.O., and H.G.G.; data curation: B. Moretti, B.H.S., and B. Motomochi; formal analysis: B.H.S.; funding acquisition: E.E., B. Moretti, B.H.S., L.A.R., A.C.O., and H.G.G.; investigation: E.E., B. Moretti, B.H.S., M.L., X.Z., S.V., G.V., C.L., P.P.F., C.J., and V.B.; methodology: E.E., B. Moretti, B.H.S., and L.A.R.; project administration: L.A.R., A.C.O., and H.G.G.; resources: L.A.R., A.C.O., and H.G.G.; software: B. Moretti, B.H.S., J.B., and L.A.R.; supervision: L.A.R., A.C.O., and H.G.G.; visualization: E.E., B. Moretti, B.H.S., G.V., C.L., P.P.F., C.J., and H.G.G.; writing – original draft: B.H.S. and H.G.G.; writing – review & editing: E.E., B. Moretti, B.H.S., J.B., X.Z., A.C.O., and H.G.G.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- [KEY RESOURCES TABLE](#)
- [EXPERIMENTAL MODELS AND SUBJECT DETAILS](#)
 - Ethics statement
 - Zebrafish
- [METHOD DETAILS](#)
 - Characterization of MCP background
 - Measurement of transgene copy number by qPCR
 - Light sheet imaging

- Image processing
- Nuclear segmentation and tracking
- Spot analysis
- Trace assembly
- Spot intensity uncertainty
- Anterior-posterior axis
- **QUANTIFICATION AND STATISTICAL ANALYSIS**
 - Quantification of MCP background levels and analysis of spot detection bias
 - Protein signal prediction
 - Kymograph creation and tissue-scale period measurement
 - Burst calling and calculating number of bursts per oscillation
 - Analysis of previously published smFISH data
 - Generalized bursting model with feedback

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, Peptides, and Recombinant Proteins		
SYBR Green MasterMix	ThermoFisher	Cat# 4309155
Low-melting-point agarose	Thermo Fisher Scientific	Cat# 16520100
Critical Commercial Assays		
DNeasy Blood & Tissue Kit	Qiagen	Cat# 69504
Deposited Data		
smFISH her1 mRNA count data (reanalyzed)	Zinani et al. ²⁸	EMBL Biostudies S-BSST434
Experimental Models: Organisms/Strains		
Zebrafish: Tg(her1:her1-MS2v5) (her1-MS2 reporter)	This paper	N/A
Zebrafish: Tg(ubi:MCP-mNeonGreen)	This paper	N/A
Zebrafish: Tg(h2b-mScarlet)	O'Brown et al. ⁴⁵	N/A
Zebrafish: AB wild-type	N/A	N/A
Zebrafish: TL wild-type	N/A	N/A
Oligonucleotides		
Primer her1-qPCR1_F: TCG ATT GGA CAC ATG AGA GC	This paper, Microsynth	N/A
Primer her1-qPCR1_R: GAA TGG AGG AGA GCT GCT TG	This paper, Microsynth	N/A
Primer ef1a_F: TCC ACC ACC ACC GGC CAT CT	This paper, Microsynth	N/A
Primer ef1a_R: CGT GCT GCG CCG CCA TTT T	This paper, Microsynth	N/A
Recombinant DNA		
Plasmid: her1-MS2 reporter (her1 regulatory region + her1 CDS + 24x MS2v5 + nls-mKate2)	This paper	Sequence at https://benchling.com/garcialab/f/f9Cist7M-zebrafish-ms2-paper/
Plasmid: ubi:MCP-mNeonGreen	This paper	Sequence at https://benchling.com/garcialab/f/f9Cist7M-zebrafish-ms2-paper/
Software and Algorithms		
zms2 (image-analysis pipeline)	This paper	https://github.com/GarciaLab/zms2 ; Zenodo 10.5281/zenodo.20349222
zebrafish-ms2-paper (figure/analysis code)	This paper	https://github.com/GarciaLab/zebrafish-ms2-paper ; Zenodo 10.5281/zenodo.20349184
OpenSimView (light-sheet microscope control)	Royer et al. ⁴⁶	https://github.com/royerlab/opensimview
Other		
FEP imaging tubes (2 mm inner diameter)	Zeus Inc., custom order	N/A

EXPERIMENTAL MODELS AND SUBJECT DETAILS

Ethics statement

All experiments with zebrafish were done in accordance with protocols approved by The University of California Berkeley's Animal Care and Use Committee and following standard protocols (protocol number 2018-01-10640-1) and at the EPFL fish facility, which has been accredited by the Service de la Consommation et des Affaires Vétérinaires of the canton of Vaud – Switzerland (VD-H23).

All zebrafish used in this study were embryos in the somitogenesis stage of development. Sex differentiation occurs later in zebrafish development⁶⁸ and thus was not a factor in our experiments.

Zebrafish

Transgenic zebrafish lines carrying *her1*-MS2 and MCP-mNeonGreen (Figure 2A) were generated using I-SceI meganuclease-mediated transgenesis.³⁹ Briefly, DNA constructs were co-injected with I-SceI meganuclease into the cell of one-cell stage wild-type embryos (AB for *her1*-MS2, TL for MCP-mNeonGreen). Post-injection, I-SceI enzyme activity was assayed by electrophoresis. Embryos were screened at 1 day post fertilization to confirm the fluorescence of mKate2 (a marker of the *her1*-MS2 transgene; see below) or mNeonGreen, whichever was appropriate to the injected DNA. Injected embryos were raised to adulthood and outcrossed to wild-type (AB or TL) zebrafish. If offspring exhibited fluorescence, they were selected to be raised for experiments and their parent was selected as a founder.

The MCP construct is driven by a ubiquitin promoter⁴⁰ and encodes a ten amino acid linker between the MS2 coat protein and fluorescent protein.

The *her1*-MS2 reporter transgene uses the shared regulatory region between *her1* and *her7*, along with the *her1* coding sequence, which recapitulates the endogenous cyclic gene expression pattern.²⁶ This design builds on a previously developed intermediate construct,⁴¹ itself derived from the earlier reporter (Soroldoni et al.²⁶

In the *her1*-MS2 construct, the *her7* coding sequence is replaced with nuclearly localized mKate2, a red fluorescent protein, which serves as a marker of transgenesis (and the presence of MS2) during embryo screening.⁴¹ The *her1* promoter drives the expression of the *her1* coding sequence followed by a 10 amino acid linker and 24 tandem copies of the MS2v5 sequence,⁶⁹ inserted immediately upstream of the *her1* 3'UTR, replacing the fluorescent protein used in earlier constructs. The insertion is predicted to maintain an in-frame fusion of the MS2 sequence with Her1. Control experiments assessing cyclic gene expression patterns and segment boundary formation indicated that any effects of the MS2 “protein” insertion on Her1 function are comparable to previously established reporters (Figure S3).

Sequences for all constructs used in this work can be found at <https://benchling.com/garcialab/f/f9Cist7M-zebrafish-ms2-paper/>.

Zebrafish lines were created at the Max Planck Institute of Molecular Cell Biology and Genetics in Dresden And then transferred to the UC Berkeley zebrafish facility and to the EPFL zebrafish facility, where they were raised with standard procedures.

To incorporate a stronger nuclear marker, we crossed the MCP-mNeonGreen fish with a line carrying h2b-mScarlet.⁴⁵ The resulting offspring were screened for the presence of both fluorophores on a Zeiss AxioZoom widefield microscope.

For imaging, the combined MCP-mNeonGreen; h2b-mScarlet fish were crossed with *her1*-MS2 and the resulting embryos were collected in the morning and then transferred to 19°C after they reached the shield stage. The next morning, embryos were screened for mNeonGreen and mScarlet, and transported from UC Berkeley to the CZ Biohub for light sheet imaging. The presence of the *her1*-MS2 reporter was screened visually on the light sheet via observation of spots.

METHOD DETAILS

Characterization of MCP background

A Zeiss 980 laser scanning microscope with Airyscan (Zeiss, Germany) was used for imaging coat protein lines in the absence of the MS2 reporter. The embryos which were at the 14-somite stage were mounted between a #1.5 coverslip (Epredia, USA) and glass slide (Thermo Fisher Scientific, USA), embedded in 0.8% low melt agarose (Thermo Fisher Scientific, USA) in fish system water. The agarose was contained in a plastic ring 12 mm interior diameter and 3 mm in depth, coated in silicon grease to prevent agarose leakage.

The following imaging settings were used: Frame size 2048x2048 pixels, Water immersion objective: LD LCI Plan-Aprochromat 40x/1.2 Imm Korr DIC M27, Speed 9 (pixel dwell time 0.26 μ s), Bidirectional scanning: On, 16 bit depth, Averaging 2, zoom 1.0 (pixel size 0.104 μ m in x and y), the eGFP channel was imaged with 488 laser power 10% and pinhole 1 AU/37 microns, and Gain 850. The detector was set to a window spanning 490 nm to 561 nm. A z-stack spanning approximately 60 μ m was taken with a step size of 1 μ m.

Measurement of transgene copy number by qPCR

DNA extraction

A subset of transgenic embryos was split into 3 groups of 3 embryos each. Wildtype siblings were split in the same way. An additional group of 10 wildtype siblings was also separated to be used as calibration curves for DNA concentration and to assess primer efficiency. Genomic DNA was extracted from each group of embryos using the DNeasy Blood & Tissue Kit (Qiagen) following the manufacturer's instructions, but resuspending the DNA in 75 μ l of milliQ water instead of the suggested 200 μ l of TE buffer. An RNase treatment was done during the extraction. Extracted DNA was stored at -25 °C until the moment of use as substrate for quantitative real-time PCR (qPCR).

qPCR

Extracted DNA was prepared to be used as template for qPCR amplification using the SYBR Green MasterMix (ThermoFisher) following the manufacturer's instructions. Primers targeting *her1* and a reference gene (eukaryotic translation elongation factor 1

alpha 1, *ef1a*) were synthesized (Microsynth). The primer sequences were as follows: *her1*-qPCR1_F: TCG ATT GGA CAC ATG AGA GC, *her1*-qPCR1_R: GAA TGG AGG AGA GCT GCT TG, *ef1a*_F: TCC ACC ACC ACC GGC CAT CT, *ef1a*_R: CGT GCT GCG CCG CCA TTT T. The qPCR was performed in an Applied Biosystems® QuantStudio 7 (ThermoFisher). Each DNA extraction, including the serial dilution of wildtype embryos, was assessed by triplicate. The qPCR program used was: 50°C (2 min), 95°C (10 min) and 36 cycles of 95°C (15 s), 72°C (1 min). In all cases the temperature gradient was 1.6 °C/s.

Estimation of the copy number

The copy number of the *her1*-MS2 transgene was calculated using the relative quantification method proposed by Pfaffl.⁷⁰ Primer efficiencies for *her1* (target gene) and *ef1a* (reference gene) were determined from standard curves generated using serial dilutions of DNA extracted from the pool of 10 wildtype embryos. For each biological replicate, the relative gene dosage of the target gene was calculated by doing the ratio of the efficiency-adjusted Ct values of the target and reference genes. These normalized ratios were then averaged across biological replicates for each genotype. The fold change in relative target gene abundance between transgenic and wild-type samples was used to estimate transgene copy number, under the assumption that each gene copy contributes equally to the qPCR signal: Transgene copies = $2 * \text{Fold Change} - 2$. With this method, the copy number of the transgene *her1*-MS2 was estimated to be 6 ± 1 (mean $\pm$ SEM). In all cases, the Ct values were the average of 3 technical replicates.

Light sheet imaging

Embryos were imaged using OpenSimView, a 4-objective multiview home-made light sheet microscope.⁴⁶ Embryos were dechorionated using sharp forceps, embedded in 0.1% low melting point agarose (dissolved in E3 media), and pipetted inside optically clear FEP tubes (Zeus Inc. custom order, 2 mm inner diameter) following an established protocol.⁷¹ E3 media was defined as 5mM NaCl, 0.17mM KCl, 0.33mM CaCl₂, 0.33mM MgSO₄, pH adjusted to 7.3 with Tris-HCl pH 7.5 1M. A plug of 1.5% agarose was inserted at the bottom of the tube to prevent the leakage of the lower-concentrated agarose. The embryo was oriented such that their tailbud pointed toward the wall of the tube. Once mounted on the microscope, the samples were excited with two light sheets using 488 nm and 560 nm lasers sequentially.

Initial assessment of our ability to detect *her1*-MS2 fluorescent spots was done using a Zeiss Z.1 Light Sheet Fluorescence Microscope (Figure S4A), which demonstrated that spots can be readily detected on a standard commercial light sheet fluorescence microscope. However, for time lapse imaging we used our custom 4-objective microscope, which produced higher signal:noise ratios and also was the source of the training data for our nuclear tracking algorithm, which resulted in more accurate tracking.

Multiview light sheet details

The light sheets were generated using two Nikon N10XW-PF objectives (0.3 NA, 3.5 mm WD). Stacks of 400 z-slices with a spacing of 0.97 μm and a frame interval of 1 minute were acquired during 3 hours at 28 °C. The total size of the field of view was 388x1000x1000 μm (z-y-x), with a pixel size of 0.485 μm in x-y. Images were acquired using two Nikon N16XLWD-PF detection objectives (0.8 NA, 3.0 mm WD) and two Hamamatsu ORCA-Flash 4.0 V3 Digital CMOS cameras. The use of two lightsheets and two cameras results in four images per channel for each time point, which are later computationally fused as described below.

During image acquisition, we encountered a hardware communication issue that led to a blocked filter wheel and subsequently missed images for approximately 10% of time points. We removed these blank time points but accounted for the missing time intervals. The rest of the scans were unaffected.

Image processing

The raw images were first converted to the *zarr* format.⁷² This format divides the image data into smaller tiles (i.e., chunks) that are compressed on the disk and are loaded on demand, allowing faster data loading and avoiding excessive memory compared to TIFF. The *zarr* format also addresses the problem of concurrent data access present in the *h5* file format.⁷² Then, for each time point, the four images that constitute each pair of light sheets and cameras were fused using the *dexp* Python package⁷³ as done in.⁷⁴ In brief, the fusion is done in two steps. First, the volumes from different light sheets and the same camera are fused, resulting in one volume per camera, two volumes in total. These two volumes are registered to compensate for a minimal misalignment between the cameras and then fused, resulting in a single volume. During fusion, small differences in the overall brightnesses of the two camera views are corrected for by linearly rescaling the dimmer image to best match the 1% and 99.9% quantiles of the brighter image. After fusing the images, the intensity of the Histone channel was normalized across the time series. A Gaussian blur and background subtraction were also applied during this step to facilitate nuclear segmentation, but no further processing was done to the MCP channel.

Nuclear segmentation and tracking

Cell nuclei and their boundaries were detected using a 3D U-NET.^{75,76} Detected nuclei were segmented and tracked over time using the approach described in^{76,77} using the following parameters: *threshold*=0.25, *max_area*=7500, *min_area*=200, and *min_frontier*=0.0. Both the segmentation and tracking were performed on a desktop computer (Intel Core i9 11900K, GeForce RTX 3090 24GB, 128 GB RAM) running Ubuntu 20.04.

Spot analysis

To perform spot analysis, we developed a custom Python pipeline. Our pipeline takes in two *zarr* arrays, one corresponding to the MCP channel images and the other corresponding to a label matrix that encodes the nuclear segmentation and tracking information, and outputs a *pandas* DataFrame with spot location and intensity information.

The goal of the spot analysis is to locate all of the true MS2 spots and quantify their intensity. The pipeline is organized into the following steps (Figure S5A):

1. Detection

A first pass of detecting spots is performed using simple filtering and thresholding.

2. Classification

Voxels containing detected spot-like objects are classified into ‘spot’ and ‘not spot’ using a convolutional neural network classifier which outputs a probability of belonging to the ‘spot’ class. The user provides a threshold probability value to reject false spots (we used 0.7).

3. Quantification

The intensity of the remaining spots is measured by estimating a background level of pixels just outside the spot, subtracting this background from spot pixels, and summing the resulting pixel values.

Detection

First, bright skin cells that surround the presomitic mesoderm were removed from the image using blurring and thresholding. Then, potential spots were identified with a simple Difference of Gaussians filter and thresholding the filtered image, with parameters chosen by visual inspection of a single time point. The filtering and binary thresholding is done on each 2D z-slice independently on a GPU using the *cucim* library. The resulting 3D mask was then transferred back to the CPU for the label matrix calculation. The centroid of each object in the label matrix was computed as an approximate location of the potential spots in the image.

Classification

A small convolutional neural network was trained to classify voxels as containing a spot or no spot. Voxels were 9x11x11 pixels³. The network was implemented in *Keras*. The network architecture was inspired by a model used to classify fluorescence microscopy images of bacteria.⁷⁸ Schematically, the network consists of 2 convolution layers followed by a fully connected layer that outputs the probability of containing a spot. Specifically, the first 3D convolutional layer has 4 kernels, each of size 3x3x3 pixels³, ReLU activation and ‘same’ padding, followed by max pooling. The second 3D convolutional layer has the same parameters as the first and is also followed by max pooling. We then flatten into a dense layer with 512 neurons and ReLU activation. During training, this layer is followed by a Dropout layer with dropout = 0.5. The result is then fed into a single sigmoid neuron for output.

The small size of this network allows it to be trained on a small number of hand-labeled spots. We manually classified 1250 voxels from 4 time points, resulting in a training dataset with 227 spots and 1023 “not spots”. We trained the model by minimizing the binary cross-entropy loss using the Adam optimizer with a learning rate of 10⁻⁴ and batch size of 8 in 5-fold cross validation. We augmented the training data with random rotations. To offset class imbalance during training, we used class weights equal to half the inverse fraction of each class in the training data. After 100 training epochs, the validation loss decayed to 0.21 ± 0.05 (mean ± standard deviation) and remained stable (Figure S5C). The final average validation area under the curve (AUC) was 0.97 ± 0.01.

During spot analysis, the initial set of detected candidate spots is filtered down using a probability threshold of 0.7.

Quantification

Spot quantification occurs through particle localization, background estimation and subtraction, and then integration over a defined ellipsoid of size (4x4x4) pixels³, or (2x2x4) microns³. Our pipeline contains two particle localization algorithms: least-squares Gaussian fitting and the radial center algorithm,⁷⁹ which is an analytic estimate of particle center based on radial symmetry. Both methods are highly accurate, approaching the Cramer-Rao bound.⁷⁹ The radial center algorithm is inherently much faster, as it requires no fitting. However, we implemented multi-processor parallelization of *scipy*’s *least_squares* function that is quite fast in practice. As discussed more below, Gaussian fitting has the benefit that the best-fit width parameters have strong predictive power in further discriminating between true and false spots. Throughout this paper, we used the Gaussian fitting method.

Because of the background MCP-mNeonGreen expression in and around each nucleus has spatial structure, straightforward application of both particle localization algorithms, which assume uniform background with Poisson noise, fail on the majority of spots. To minimize the structured background, we first perform Difference of Gaussians filtering on the spot voxel and then estimate particle centers.

Next, we estimate the background levels for each spot individually and subtract that background from the spot pixel intensities. Background estimation can be done in one of two ways. The first way is to re-fit a 3D Gaussian to the spot in real space, but with fixed center and width parameters, extracting just an offset and amplitude. Gaussian fitting is the standard way to estimate background levels of MS2 spots.¹¹ The second way is to take a “shell” of pixels of a fixed width and distance from the spot center and average the intensity of the shell. Using a shell of inner width = 4 pixels and outer width = 6 pixels, we find that the two methods of background estimation strongly agree with one another (Figure S5F, R² = 0.94). In all data in this paper, we used the shell method, as it is faster.

With the background estimate in hand, we subtract this background from all pixels in the voxel, enforcing non-negativity. We then define an ellipsoid of size 4x4x4 pixels³ and sum the background-subtracted intensities.

To assess the accuracy of this quantification procedure, we generated a library of synthetic spots with known integrated intensity and realistic background and noise (Figure S6A). Specifically, we modeled spots using a Gaussian point spread function (PSF) and a constant offset. We then modeled the structured background by adding to this Gaussian an exponential function with decay length equal to 10 pixels and a random orientation in 3D. Finally, we modeled shot noise by assigning to each pixel value a Poisson random number with mean equal to the intensity of the noise-less pixel.

We ran our spot quantification pipeline on this library and assessed intensity accuracy as a function of spot amplitude (which sets the signal-to-noise ratio) and the amplitude of the exponential gradient. We measured the relative accuracy of the intensity measurement, defined as (measured intensity - true intensity) / true intensity, across spot parameters and also using two different spot localization methods: first, Gaussian fitting on the raw pixels, and second, Gaussian fitting on Difference-of-Gaussians (DoG)-filtered images, the latter to remove the structured background. We found that localization using DoG-filtered images led to considerably enhanced accuracy across spot parameters (Figure S6B).

To make contact with the experimental data, we defined two readily measured parameters: signal:background ratio (where signal is the mean background-subtracted spot pixel intensity) and the “structuredness” of the background, defined as the variance-to-mean ratio of the of the shell pixels that define the background. For a uniform background with Poisson noise, the structuredness is 1. Binning our simulations in this 2D space, we created an accuracy regime diagram (Figure S6C). Averaging across all spots in our dataset, we found that the experimental data had a signal:background value of 0.35 ± 0.24 (mean $\pm$ standard deviation), i.e., 35% above background, and a structuredness value of 40 ± 21 , reflecting the highly structured MCP background visible in the spot images. Nevertheless, with DoG-based localization and our Gaussian combined with our exponential model of spots, we infer that our spot intensity quantification is highly accurate, with a relative error of at most 40%, but typically around 5%. For visual clarity, spot intensity is sometimes reported as a fraction of the brightest spot in the dataset.

Trace assembly

MS2 traces are assembled by assigning spots to nuclei and linking them through time using the nuclear tracking information. First, we check if a spot’s centroid falls within a region identified as a nucleus by the “Segments” label matrix that is output from the nuclear tracking pipeline. If a spot does not fall within a nucleus, then we perform a search for the nearest nucleus within a cube of size 11 pixels in x and y and 7 pixels in z, centered on the spot. If multiple spots are assigned to the same nucleus, we pick the one with the highest probability value from the neural network classifier.

After assigning each spot to a nucleus, we assemble a draft of the traces and begin iteratively refining them. In each iteration we perform the following steps. We identify regions of transcriptional activity by thresholding a moving average of the trace using a window of 5 time points and a threshold of 1 a.u. (effectively any non-zero point passes the threshold). For each point in the “on” region of the trace such that the raw trace has a value of zero and has an adjacent point with a value greater than zero, we attempt to fill in that point. We extract a voxel centered on the location of the adjacent point but at the “empty” time point. We then run this voxel through our spot classification and quantification routines. We accept or reject the spot using either the spot classifier or quantification parameters. For all data in this paper, we kept only spots for which all Gaussian width parameters (sigma_x, sigma_y, and sigma_z) fell within the range 0.5 and 3.0 pixels, which we found to be an effective filter by manual inspection.

Upon iterating this process, the set of found points that pass the given criteria will converge. For all data in this paper, we used 10 iterations.

After the traces are assembled, we can make further cuts by rejecting spots based on spatial location. Previous measurements of protein levels²⁹ and smFISH counts⁵³ indicate that *her1* transcription halts after somite boundary formation. For reasons unknown, we found that background levels of MCP-mNeonGreen increase in somites after boundary formation, leading to increased false positive detections in this region. We therefore reject spots that are greater than 40 μ m anterior of the last formed somite, as measured along our defined anterior-posterior axis (see below). The position of the last formed somite was determined manually over time. Similarly, we restrict our analysis to spots detected within the PSM and tailbud tissues by rejecting spots that lie greater than 50 μ m from the anterior-posterior axis (see below), corresponding to, for example, false detections in the skin. Finally, for analysis of single-cell traces, we keep only traces with greater than 10 spots.

Spot intensity uncertainty

The uncertainty in individual spot intensity was estimated using a method based on.¹¹ A full derivation of the uncertainty is given in Method S2. In brief, the uncertainty was assumed to be dominated by the error in background estimation. This error is measured by considering the time evolution of the background around each spot, fitting a mean trend using a 4th order polynomial, and computing the RMS deviation of background levels from this trend. In contrast to the original method,¹¹ we use a 4th order polynomial as opposed to a spline as we found that additional degrees of freedom in the trend model led to worse performance in a cross-validation scheme (Method S2; Figure S20).

In addition, and in contrast to previous works, due to the increased difficulty of the spot detection problem we further incorporate uncertainty in spot detection via the empirically measured false-positive and false-negative rates (Method S2.2). Combining both of these sources of uncertainty, we arrive at the final uncertainty for an MS2 traces as

$$\sigma = \sqrt{(\sigma_l)^2(1 - f_p) + I^2 f_p(1 - f_n)}, I > 0 \quad (\text{Equation 1})$$

$$\sigma = \bar{I} \sqrt{f_n(1 - f_n)}, I = 0$$

where σ_l is the uncertainty from background fluctuations, f_p is the false-positive rate of detection, f_n is the false-negative rate of detection, I is the intensity of the individual spot, and $\bar{I}$ is the mean intensity of the trace. From analysis of manually-curated traces, we measured $f_p = 0.04$ and $f_n = 0.08$. σ_l depends on the trace, but is typically 1%–10% of $\bar{I}$. See Method S2 for more details.

Anterior-posterior axis

We define the anterior-posterior axis using a combination of manually placed points and spline interpolation. Using the interactive data viewer *napari*,⁸⁰ we construct a coarse anterior-posterior axis by manually placing around 10 points that span from the last formed somite (which becomes the origin) to the tip of the tail bud in the first time point of the timeseries. We place these points in 3D using a *napari* Points layer, viewing one slice at a time, roughly following the notochord. In the tailbud, we complete the anterior-posterior axis by projecting points towards the closest outer boundary of the tissue. Keeping track of the location of that last formed somite, we proceed to define coarse anterior-posterior axes every 50 time points until the end of the Video. We then refine these anterior-posterior axes through spline interpolation. First, we interpolate the anterior-posterior axis of each time point, going from 10 to 100 spatial positions, using a B spline of degree 2. Then, for each 100 points along the new anterior-posterior axis, we interpolate over time, using a B spline of degree 1. We use *scipy*'s *splprep* function to define the spline. Spots are assigned to an anterior-posterior axis bin by finding the axis point that has the smallest Euclidean distance to the spot's centroid.

QUANTIFICATION AND STATISTICAL ANALYSIS

Quantification of MCP background levels and analysis of spot detection bias

To assess how heterogeneity in MCP background fluorescence might bias spot detection, we quantified the average MCP intensity per nucleus throughout the Video. For this, we used the nuclear segmentation and tracking output to define cell boundaries, and for each cell and time point computed the mean MCP-mNeonGreen intensity within that region (using *skimage.measure.regionprops* in Python). The resulting values were corrected for photobleaching by normalizing each time point by a bleaching correction factor. This correction was obtained by fitting an exponential decay to the mean MCP intensity over all non-zero pixels in each image frame.

To restrict the analysis to regions of the embryo relevant to *her1* expression (i.e., the presomitic mesoderm and tailbud), we considered only nuclei posterior to the last formed somite at each time point, as determined from the manually defined somite boundary positions used throughout the study. Bright skin regions were removed using morphological filters, but some residual skin cells remained and were included in the analysis.

We then compiled the mean MCP intensities from all tracked nuclei across 162 time points to generate histograms representing (i) all nuclei after filtering (Figure S7A, blue histogram, $n=31,718$ cells), and (ii) only nuclei that displayed MS2 spots during the Video (Figure S7A, red histogram, $n=403$ cells). Both histograms overlapped across most of their ranges, indicating that spot detection was not substantially biased by heterogeneous MCP levels. Cells with the very highest MCP intensities corresponded to unfiltered skin nuclei and lacked MS2 spots. An overview of the spatial heterogeneity of MCP background fluorescence is shown in Supplementary Figure 7B.

Protein signal prediction

We use a linear model of transcription and translation to predict what mRNA (m) and protein (p) concentrations, measured in arbitrary units, would be created by our MS2 traces, $r_m(t)$, assuming that the fluorescent intensity of the MS2 signal is proportional to the instantaneous rate of transcription.⁴⁸ Specifically, we model the system according to

$$\frac{dm}{dt} = r_m(t) - \gamma_m m \quad (\text{Equation 2})$$

$$\frac{dp}{dt} = r_p m - \gamma_p p \quad (\text{Equation 3})$$

When no spot is detected, the MS2 trace $r_m(t)$ is assigned a value of zero. We numerically integrate this model using *scipy*'s *solve_ivp* function with the RK45 routine and interpolating the MS2 signal, $r_m(t)$, between observation times using *numpy*'s *interp* function.

The key parameters are the decay rates (γ_m, γ_p) of *her1* mRNA and Her1 protein, respectively, which we take from reference⁵⁸ to both be 0.23 min^{-1} . The production rate constants for mRNA and protein simply set the concentration scales of mRNA and protein, respectively, which here can be taken as arbitrary since we measure mRNA levels and predict protein levels in arbitrary units. The initial conditions of total mRNA and protein are unknown parameters. In Method S1, we show that the predicted mRNA and protein signals become independent of the initial condition on the time scale of the decay constants. Therefore, for simplicity, we pick initial conditions of $m(t=0)=\text{protein}(t=0)=0$ and ignore approximately the first cycle of predicted signals.

In Figures 1C–1E, we use this model to demonstrate the low-pass filtering effect of the central dogma (more details in Method S1). We generated toy example MS2 traces consisting of either a sine wave or a square wave and used these as inputs to the protein prediction model, which outputs smooth oscillations in both cases.

Kymograph creation and tissue-scale period measurement

To create a kymograph, spots were grouped by their anterior-posterior position into 100 equally spaced bins (approximately $5 \mu\text{m}$ long). The intensities were summed within each bin and plotted as a heatmap with time increasing downwards. The lengthening of the tissue-scale period was measured directly from the kymograph by summing the intensities from bins located between 240 and $315 \mu\text{m}$ along the anterior-posterior axis and plotting the resulting signal over time. Oscillation peaks were identified by fitting

a Gaussian profile to manually-defined sections of the signal that isolated single oscillations and taking the fit center. Periods were calculated as the differences between successive peaks. Uncertainty in the period was obtained by bootstrapping over spatial bins within the 240–315 μm range. The predicted protein measurements were done identically, but instead of the spot intensities using the simulated protein signals evaluated at the observation time points.

The kymograph schematic in Figure 4B was created with the “Phase-Amplitude” model described in⁸¹ and the Python code contained therein. In brief, the model consists of a 1D grid of phase oscillators with spatially varying frequencies and exponentially growing amplitudes. When the oscillating signals exceed a threshold value, the oscillations cease, representing somite formation.

Burst calling and calculating number of bursts per oscillation

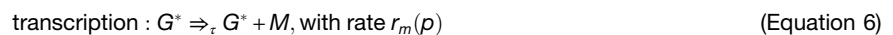
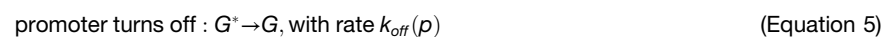
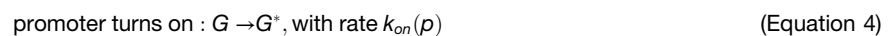
We implemented a conservative burst calling algorithm based on thresholding a moving average of the *her1*-MS2 traces. The moving average was computed via convolution with a uniform kernel of size 3 time points, and we used a threshold of 1 a.u., effectively grouping any non-zero signal into a burst. With this algorithm, fluctuations in *her1* transcription are determined as corresponding to distinct bursts only if they are separated by 3 time points of inactivity. We defined oscillations using the predicted protein signal. We used *scipy*’s *find_peaks* function to identify peaks in the predicted protein oscillation. We then collected burst start and end times and assigned the bursts starting between the previous protein peak and the current protein peak to the current protein peak. With all bursts being assigned to a protein peak, we then computed a histogram of the number of bursts per protein peak, which we call the number of bursts per oscillation. With our conservative burst calling algorithm, we likely underestimate the frequency of stochastic bursting within individual Her1 protein oscillations.

Analysis of previously published smFISH data

Counts of *her1* mRNAs per cell across 23 embryos grown at 28°C were downloaded from Table S6 of Zinani et al.²⁸ We fit the analytic solution for the stationary distribution of mRNA counts from the canonical two-state model of transcriptional bursting⁸² to each embryo’s counts using maximum likelihood estimation. Specifically, we minimized the negative log-likelihood, using *scipy*’s *loggamma* function to avoid numerical blow up. An example raw image of the smFISH experiments was downloaded from the public EMBL Bio-studies repository with accession number S-BSST434.

Generalized bursting model with feedback

Details of the mathematical model and its implementation are found in Method S3. In brief, we extended the canonical two-state bursting model to allow for protein concentration to increase the rate of transcription, decrease the rate of OFF $\rightarrow$ ON events, or increase the rate of ON $\rightarrow$ OFF events (Figure 6A) according to Hill functions. The model contains three chemical species: the *her1* gene, which can be in the inactive state, G , or the inactive state, G^* ; *her1* mRNA, denoted by M ; and Her1 protein, P , whose total number is denoted by p . These three species are governed by the following set of reactions:



We performed fully stochastic Gillespie simulations of the model, with the mRNA production reaction occurring with a time delay, τ , (denoted by the double arrow with a τ subscript) following Bratsun et al.⁸³

Model parameters

The following parameters were held fixed: mRNA decay rate (γ_m) = 0.23 1/min, protein decay rate (γ_p) = 0.23 1/min,⁵⁸ maximum transcription rate (r_m) = 10 mRNAs/min (chosen to obtain an oscillation amplitude of approximately 40 mRNAs,⁵³ translation rate (r_p) = 4.5 proteins/mRNA/min. The remaining model parameters were chosen manually to produce simulated MS2 traces that qualitatively resemble the experimental data and exhibit statistically increased regularity in the resulting oscillations compared to unregulated bursting. For the simulations in Figures 6 and S11–S17, the following parameters were used:

- Amplitude regulation: the transcription rate is repressed by the protein according to a Hill function, $r_m/(1 + (p/K_D)^n)$. We ignore promoter switching (i.e., the promoter is always in the ON state by setting $k_{off}=0$ 1/min), letting the sharp repression of the transcription rate itself create burst-like behavior. Parameters: $K_D=100$ proteins, $n=3$, $\tau=7.5$ min.
- Separation regulation: k_{on} is repressed according to a Hill function, $k_{on} = k^+/(1 + (p/K_D)^n)$, with a maximum value of k^+ . Parameters: $k^+=0.5$ 1/min, $k_{off}=0.08$ 1/min, $K_D=80$ proteins, $n=3$, $\tau=0$ min.

- Duration regulation: k_{off} increases with protein level according to an increasing Hill function, $k_{\text{off}}(p) = k^-(p/K_D)^n / (1 + (p/K_D)^n)$, with a maximum value of k^- . Parameters: $k_{\text{on}}=0.055$ 1/min, $k^- = 0.4$ 1/min, $K_D=1100$ proteins, $n=3$, $\tau=0$ min.
- Separation and duration regulation: both k_{on} and k_{off} are Hill functions of protein concentration, with maximum values k^+ and k^- , respectively. Parameters: $k^+=0.5$ 1/min, $k^- = 0.4$ 1/min, $K_{D,\text{on}}=80$ proteins, $K_{D,\text{off}}=1100$ proteins, $n=3$, $\tau=0$ min.
- Amplitude and duration regulation: both k_{off} and r_m are regulated by Hill functions with different K_D values. The parameters from the duration regulation simulations were used together with the transcription rate parameters from the amplitude regulation simulations.
- Amplitude and separation regulation: both k_{on} and r_m are regulated by Hill functions with different K_D values. The parameters from the separation regulation simulations were used together with the transcription rate parameters from the amplitude regulation simulations.
- Amplitude, duration, and separation regulation: all three parameters are regulated by Hill functions, combining the respective parameters from the three simulations.
- Unregulated bursting: no burst parameters are regulated, resulting in a traditional two-state bursting model with $k_{\text{off}}=0.08$ 1/min, $k_{\text{on}}=0.055$ 1/min, and $r_m=10$ mRNAs/min.
- Stochastic transcription rate regulation: here, we set $k_{\text{off}}=0.0$ 1/min, such that the promoter is always in the ON state, and have the transcription rate regulated by a Hill function as in the amplitude regulation case, with a delay of $\tau=7.5$ min.
- Deterministic transcription rate regulation: here, we solve a deterministic delay differential equation that corresponds to the deterministic limit of the stochastic transcription rate regulation model above. Equations for this model are given in [Method S3](#), equations (48,49).

Simulated MS2 signals

From each simulation we extracted a time series of the promoter state variable, which fluctuates between 0 (“OFF”) and 1 (“ON”). Using these promoter state traces, we simulated realistic MS2 traces using the model of⁴⁸ that encapsulates the effect of RNA polymerases being loaded on and off the gene with a memory kernel of normalized length w corresponding to the dwell time of RNA polymerase on the gene as it engages in transcription. Based on the measured elongation rate for *her1* via smFISH⁸⁴ 4.8 kb/min and our approximately 1.4kb MS2 cassette at the 3’ end of the transgene, we used a dwell time = 0.29 min, which, with our sampling rate of 1.0 1/min implies $w = 0.29$. We further included multiplicative Gaussian noise with standard deviation σ_{MS2} to simulate fluorophore emission and measurement uncertainty, and a detection threshold, below which the simulated MS2 signal was set to zero. We used default values of $\sigma_{\text{MS2}} = 0.2$, and detection threshold = $0.2 * w * \max(r_m)$, where r_m is the mRNA production rate in [equation 6](#) and is time-dependent in the case of amplitude regulation. These values were chosen because they generated traces with a scale of fluctuations that visually matched our experimental data. However, we also varied these parameters in our robustness analysis ([Figures S12–S15](#)).

Protein period regularity measurement

To assess the regularity of simulated protein oscillations in our model, we used the *scipy* find_peaks function with prominence = 0.01 to identify the locations of peaks in protein oscillations. We then defined the protein period as the intervals between peaks and computed the coefficient of variation (CV, standard deviation/mean) of these periods.